Math 871 Topology I
Problem Set 8

1. Consider the box topology on $\mathbb{R}^\omega$, and let A be the subset of $\mathbb{R}^\omega$ consisting of all the sequences whose terms are strictly positive:

$$A = \{x = (x_n)_{n\geqslant 1} \in \mathbb{R}^\omega \mid x_n > 0 \text{ for all } n \geqslant 1\}.$$

   (a) Let $z = (0)_{n\geqslant 1} \in \mathbb{R}^\omega$ be the constant sequence equal to zero. Show that $z \in \overline{A}$.

   (b) Show that there is no sequence of elements of A converging to z.
   Hint: given any sequence $(a_n)_{n\geqslant 1}$ of elements of A, let

$$a_n = (x_{n,1}, x_{n,2}, x_{n,3}, ...) = (x_{n,i})_{i\geqslant 1}$$

   and consider the open set

$$U = \prod_{n=1}^{\infty} (-x_{n,n}, x_{n,n}).$$

   (c) Conclude that $\mathbb{R}^\omega$ with the box topology is not metrizable.

2. Let $\overline{\rho}$ be the uniform metric on $\mathbb{R}^\omega$. Given $x = (x_n)_{n\geqslant 1} \in \mathbb{R}^\omega$ and $0 < \varepsilon \leqslant 1$, let

$$U(x, \varepsilon) = \prod_{n=1}^{\infty} (x_n - \varepsilon, x_n + \varepsilon).$$

   Let $z = (0)_{n\geqslant 1} \in \mathbb{R}^\omega$ be the constant sequence equal to zero. Show that $U(z, 1)$ is not open in the uniform topology.